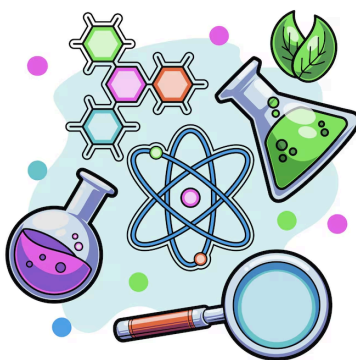




UNIVERSITY INTERSCHOLASTIC LEAGUE

Science

Invitational B • 2026



GENERAL DIRECTIONS:

- **DO NOT OPEN EXAM UNTIL TOLD TO DO SO.**
- Contestants may take up to two hours to complete the contest. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
- Papers may not be turned in until 30 minutes have elapsed. If you finish the test in less than 30 minutes, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
- All answers must be written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet. Write clearly and legibly!
- You may place as many notations as you desire anywhere on the test paper but not on the answer sheet, which is reserved for answers only.
- You may use additional scratch paper provided by the contest director.
- All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers.
- If a question is omitted, no points are given or subtracted.
- The back two pages of this test include a copy of the periodic table of the elements, as well as listings of other scientific relationships. You may use this information during the contest and may detach the back page from the test if you wish.
- A simple scientific calculator is sufficient for the high school Science contest. **The UIL provides a list of approved calculators that meet the criteria for use in the Science contest. No other calculators are permitted during the contest.** The Science Contest Approved Calculator List is available in the current Science Contest Handbook and on the UIL website. Contest directors will perform a brief visual inspection to confirm that all contestants are using only approved calculators. Each contestant may use up to two approved calculators during the contest.

- B01. In incomplete dominance, what is observed in the heterozygous phenotype?
- A) Only the dominant trait appears.
 - B) Both alleles are expressed fully and separately.
 - C) The phenotype is a blend of the two alleles.
 - D) The recessive allele is silenced by methylation.
 - E) The heterozygous individual is sterile.

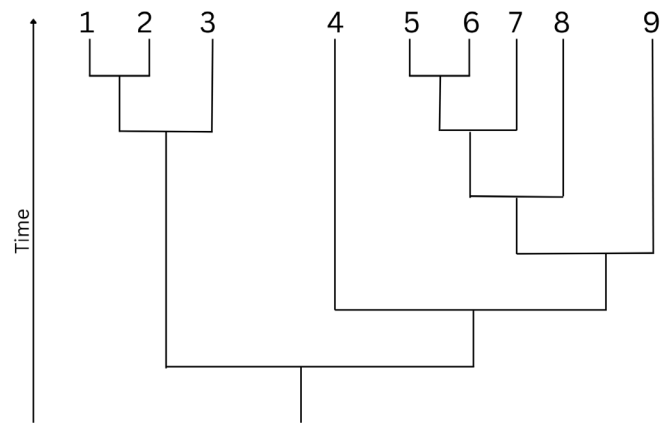
- B02. Which of the following is an ecological role of bacteria?
- A) Primary producers
 - B) Pathogens
 - C) Decomposers
 - D) Nitrogen fixation
 - E) All the above

- B03. Which of the following regulates progression through cell cycle checkpoints such as G1 and G2?
- A) Histone acetylation and DNA methylation.
 - B) Cyclins and cyclin-dependent kinases (CDKs).
 - C) Spindle fiber length and centrosome duplication.
 - D) Telomerase activity and ribosome number.
 - E) Mitochondrial membrane potential and oxygen availability.

- B04. In an experiment, a group of cells placed in a hypertonic solution shrivel and shrink. Which cellular structure is most responsible for controlling water movement into and out of the cell?
- A) Plasma membrane
 - B) Mitochondrial membrane
 - C) Nuclear envelope
 - D) Endoplasmic reticulum
 - E) Cell wall

- B05. Which of the following best describes a homologous structure?
- A) Traits that serve the same function but have different evolutionary origins.
 - B) Traits that appear identical in all species.
 - C) Anatomical features that are inherited from a common ancestor.
 - D) Traits that arise due to similar environmental pressures in unrelated species.
 - E) Fossils with no clear modern counterparts.

- B06. Examine the phylogenetic tree below. Which species shares a more recent common ancestor with species 7?



- A) 1
- B) 3
- C) 4
- D) 8
- E) 9

- B07. What is the primary mode of transmission for *Bordetella pertussis*?
- A) Respiratory droplets
 - B) Contaminated food and water
 - C) Airborne spores
 - D) Bloodborne exposure
 - E) Contact with contaminated surfaces

- B08. In guinea pigs, black fur (B) is dominant over white fur (b). Two heterozygous black guinea pigs are crossed. Out of 500 offspring, approximately how many would you expect to have white fur?
- A) 0
 - B) 125
 - C) 250
 - D) 375
 - E) 500
- B09. Which of the following plant responses is regulated by phototropism?
- A) The closing of stomata during a drought.
 - B) The movement of chloroplasts toward the cell wall.
 - C) The growth of a plant stem toward light.
 - D) The downward growth of roots in soil.
 - E) The opening of flowers at night.
- B10. Which organ system protects the body from external pathogens and helps regulate body temperature?
- A) Respiratory
 - B) Circulatory
 - C) Digestive
 - D) Nervous
 - E) Integumentary
- B11. A researcher discovers a molecule made of carbon, hydrogen, and oxygen in a 1:2:1 ratio. It is readily soluble in water and breaks down to release energy. This molecule is most likely a
- A) monosaccharide.
 - B) steroid.
 - C) polypeptide.
 - D) nucleic acid.
 - E) phospholipid.
- B12. In November 2025, the Center for Disease Control and Prevention and public health officials investigated an ongoing multistate outbreak of _____ linked to infant formula.
- A) *Salmonella*
 - B) Hepatitis B
 - C) *Listeria*
 - D) *E. coli*
 - E) botulism
- B13. During cellular respiration in eukaryotes, the electron transport chain and ATP synthase are embedded within which structure, allowing chemiosmosis to occur across it?
- A) Chloroplast outer membrane
 - B) Thylakoid membrane
 - C) Mitochondrial matrix
 - D) Inner mitochondrial membrane
 - E) Nuclear envelope
- B14. A competitive inhibitor binds to the site on an enzyme where the substrate normally attaches. How would this affect the enzyme's function?
- A) It blocks substrate binding by occupying the active site.
 - B) It changes the enzyme's shape, preventing substrate binding at a different site.
 - C) It increases enzyme activity by stabilizing the substrate.
 - D) It permanently deactivates the enzyme by destroying the active site.
 - E) It binds the substrate, enhancing the reaction rate.
- B15. Which of the following best describes an operon that is normally inactive, but can be turned on in the presence of a specific substrate?
- A) Repressible operon
 - B) Inducible operon
 - C) Promoter
 - D) Operator
 - E) Constitutive operon

- B16. A small group of birds is blown off course onto a new island, where they establish a population. Over time, this population shows reduced genetic variation compared to the mainland. This is best explained by:
- A) Stabilizing selection
 - B) Founder effect
 - C) Mutation
 - D) Gene flow
 - E) Overdominance
- B17. What does population density refer to?
- A) The number of individuals per unit area.
 - B) The total number of individuals in a population.
 - C) The genetic diversity within a population.
 - D) The rate at which a population grows over time.
 - E) The number of species in a community.
- B18. Negative feedback systems are the primary way that the body maintains stability, known as ____.
- A) metabolism
 - B) growth
 - C) reproduction
 - D) differentiation
 - E) homeostasis
- B19. Which of the following best describes the lytic cycle of a virus?
- A) The virus integrates its genome into the host's DNA and remains dormant.
 - B) The virus hijacks host machinery to produce new viral particles and bursts the host cell.
 - C) The virus replicates inside the host without causing cell death.
 - D) The virus causes the host cell to divide uncontrollably, forming a tumor.
 - E) The virus triggers the host immune system to destroy infected cells before replication.
- B20. What happens immediately after the ribosome encounters a stop codon during translation?
- A) A tRNA recognizes the stop codon and adds the stop amino acid.
 - B) The ribosome recruits RNA polymerase to restart translation.
 - C) The ribosome detaches and releases the completed polypeptide.
 - D) The mRNA is degraded and recycled.
 - E) The ribosome begins transcribing the next gene.

C01. How many atoms are there in one formula unit of potassium carbonate?

- A) 3 B) 4 C) 5 D) 6 E) 7

C02. How many oxygen molecules are needed when the equation for the formation of iron(III) oxide from elemental iron and oxygen is balanced using the smallest whole numbers?

- A) 2 B) 3 C) 4 D) 5 E) 6

C03. What is the difference, if any, between atomic mass and mass number?

- A) Atomic mass and mass number are different terms for the same thing.
 B) Mass number is an obsolete term for atomic mass and is no longer used.
 C) Atomic mass is the decimal value at the bottom of a periodic table entry. Mass number is the integer at the top of the periodic table entry.
 D) Atomic mass is specifically the mass of an atom while mass number can be the mass of either an atom or a molecule.
 E) Atomic mass is the average mass of all isotopes of an element while mass number is the number of nucleons in one specific isotope of that element.

C04. How many of these molecules contain π bonds?

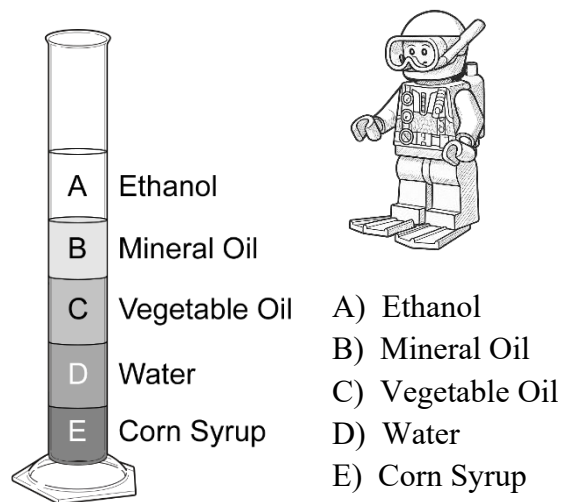
CO₂ CH₄ H₂O NH₃ CO

- A) 1 B) 2 C) 3 D) 4 E) 5

C05. If 22.4 grams of dry ice (frozen carbon dioxide) is allowed to sublime in a flexible-walled container to 25°C at 1.0 atm pressure, what will the volume of the gas be?

- A) 1.0 L
 B) 6.2 L
 C) 12.4 L
 D) 18.6 L
 E) 22.4 L

C06. A graduated cylinder is filled with samples of five different liquids. A Lego SCUBA diver is dropped into the cylinder and begins to sink. Which is the lowest layer that the diver will sink to the bottom of?



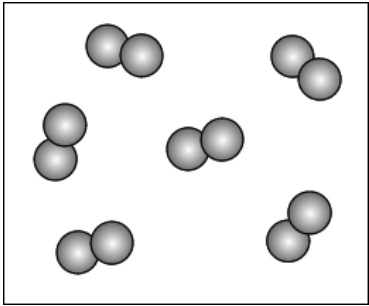
C07. The ΔH for the reaction between HCl and sodium bicarbonate is 28.5 kJ/mol. This reaction is best described as

- A) endothermic
 B) exothermic
 C) ectothermic
 D) isothermic
 E) xenomorphic

C08. In which of these chemical reactions is the carbon atom being oxidized?

- (i) $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
 (ii) $\text{CH}_4 + 4\text{Cl}_2 \rightarrow \text{CCl}_4 + 4\text{HCl}$
 (iii) $2\text{CO} + \text{O}_2 \rightarrow 2\text{CO}_2$
 (iv) $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3$

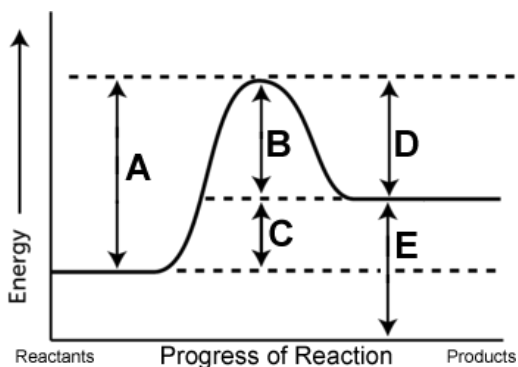
- A) i and iii
 B) ii and iv
 C) i, ii, and iii
 D) i, ii, and iv
 E) all of them

- C09. Which statement below about phase changes of water is true?
- A) Melting 10 g of ice requires as much heat as is given off when 10 g of water freezes.
 - B) It takes more heat to melt 10 g of ice to water than it takes to boil 10 g of water to vapor.
 - C) It takes the same amount of heat to melt 10 g of ice to water as it does to boil 10 g of water to vapor.
 - D) Freezing liquid water to ice absorbs heat from the surroundings, which is why ice makes things cold.
 - E) Boiling requires more energy than freezing because the H_2O molecules have to be broken into H_2 and O_2 .
- C10. Which of the changes below will change the value for the equilibrium constant K for a reaction?
- A) Increasing the pressure
 - B) Raising the temperature
 - C) Adding more reactant
 - D) Adding more product
 - E) All of these
- C11. While doing your chemistry homework you need to know the dissociation constant (K_a) for hypochlorous acid, but the Wikipedia article for hypochlorous acid doesn't include the K_a . Instead it lists the $\text{p}K_a$ as 7.53. How would you determine the K_a for this acid from $\text{p}K_a$?
- A) $K_a = \text{p}K_a$
 - B) $K_a = -\log(\text{p}K_a)$
 - C) $K_a = -\ln(\text{p}K_a)$
 - D) $K_a = e^{\text{p}K_a}$
 - E) $K_a = 10^{-\text{p}K_a}$
- C12. Which of these situations is an example of the common ion effect affecting the solubility of the added salt?
- A) Adding solid NaCl to a 0.10 M solution of NaCl
 - B) Adding solid AgCl to a saturated solution of AgCl
 - C) Adding solid AgCl to a 0.10 M solution of NaCl
 - D) Adding solid NaCl to a saturated solution of AgCl
 - E) All of these
- C13. If you want to determine how much of a reactant remains after a chemical reaction is allowed to run for a period of time, which of these would you use to calculate it?
- A) the equilibrium expression for the reaction
 - B) Kinderhook's Law
 - C) the rate law for the reaction
 - D) the integrated rate law for the reaction
 - E) the Arrhenius equation
- C14. Which term below best describes the gas sample shown here?
- 
- A) Molecules
 - B) A compound
 - C) A mixture of atoms
 - D) A mixture of compounds
 - E) A mixture of elements

C15. What is the pH of a 175 mL solution of 0.75 M NH_3 ? K_b for $\text{NH}_3 = 4.75 \times 10^{-5}$.

- A) 11.78
- B) 2.59
- C) 13.12
- D) 12.33
- E) 12.65

C16. In the potential energy diagram shown below, which arrow or sum of arrows tells you whether the reaction is endothermic or exothermic?



- A) A
- B) B
- C) C
- D) B+C
- E) B+E

C17. When solid calcium carbonate is heated, the products are carbon dioxide and calcium oxide. What type of reaction is this?

- A) Decomposition
- B) Synthesis
- C) Combustion
- D) Double Replacement
- E) Single Replacement

C18. What is the percent copper by mass in CuCl_2 ?

- A) 33.3
- B) 52.7
- C) 35.8
- D) 64.2
- E) 47.3

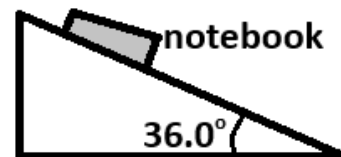
C19. What is the K_{sp} expression for the precipitation reaction between aqueous $\text{Ba}(\text{NO}_3)_2$ and aqueous NaBrO_3 ?

- A) $K_{sp} = \frac{[\text{Na}^+][\text{BrO}_3^-]}{[\text{Ba}^{2+}][\text{NO}_3^-]^2}$
- B) $K_{sp} = \frac{[\text{Ba}^{2+}][\text{NO}_3^-]}{[\text{Na}^+][\text{BrO}_3^-]}$
- C) $K_{sp} = [\text{Ba}^{2+}][\text{BrO}_3^-]^2$
- D) $K_{sp} = \frac{1}{[\text{Ba}^{2+}][\text{BrO}_3^-]^2}$
- E) $K_{sp} = \frac{[\text{Ba}^{2+}][\text{BrO}_3^-]^2}{[\text{Ba}(\text{BrO}_3)_2]}$

C20. One ampere (A) is equal to one coulomb of charge passing a given point each second. If a 2 A electrical current is passed through a solution of Ag^+ for 40 minutes, how many moles of solid Ag would be formed?

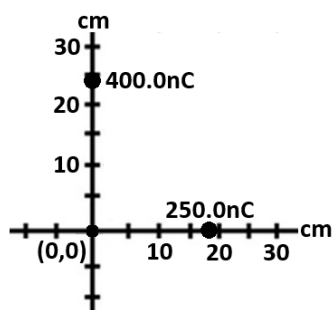
- A) 0.040 mol
- B) 0.050 mol
- C) 0.060 mol
- D) 0.070 mol
- E) 0.080 mol

- P01. According to Malley, Frederick Soddy and William Ramsay isolated the emanations from radium. In the emanation mixture, they observed the spectral lines of which element?
- Actinium
 - Polonium
 - Radon
 - Argon
 - Helium
- P02. According to Malley, Paul Villard sent radium rays through a magnetic field, which broke the rays into different parts. Which type of radiation did Villard discover by analyzing the different parts of the radium rays?
- Alpha particles
 - Beta particles
 - Gamma rays
 - Delta rays
 - Neutrinos
- P03. According to Malley, Norman Campbell and Albert Wood claimed to detect weak radioactivity from elements that are very different from uranium, radium, and thorium. Which of these is an element about which they made this claim?
- Helium
 - Lead
 - Carbon
 - Potassium
 - Germanium
- P04. Which class of main sequence stars spends the longest amount of time on the main sequence?
- Class O blue stars
 - Class A white stars
 - Class G yellow stars
 - Class K orange stars
 - Class M red stars
- P05. A leaf blowing in the wind floats horizontally across a football field (goal post to goal post, 120 yards.) The leaf takes 12.5 minutes to travel the entire distance. What is the speed of the leaf in miles per hour?
- 0.109 mph
 - 0.160 mph
 - 0.224 mph
 - 0.327 mph
 - 0.480 mph
- P06. While trying to crack a walnut, you climb up a ladder and throw the walnut straight down onto a concrete floor. You throw the walnut from a height of 2.35m, and the walnut hits the ground exactly 0.250 seconds after you throw it. With what initial velocity did you throw the walnut?
- 9.40 m/s
 - 8.18 m/s
 - 7.67 m/s
 - 5.92 m/s
 - 4.17 m/s
- P07. A 1.50kg notebook is placed on a frictionless inclined plane. The plane is angled at 36.0° with respect to horizontal, and the notebook slides down the incline as expected. While it is on the incline, what is the magnitude of the normal force (F_N) acting on the notebook?
- 14.7 N
 - 11.9 N
 - 8.64 N
 - 7.12 N
 - 5.76 N



- P08. A 600g toy cart starts at rest on a frictionless horizontal track. You throw a 350g blob of clay, which collides with the back of the cart and sticks to it. Just before the collision, the clay has a horizontal velocity of 13.0m/s. After the collision, the cart-clay system slides forward along the track. What is the speed of the cart-clay system after the collision?
- 4.79 m/s
 - 5.72 m/s
 - 6.65 m/s
 - 7.58 m/s
 - 8.21 m/s
- P09. A merry-go-round starting from rest slowly accelerates to an angular velocity of 9.00rad/s . The merry-go-round takes two complete revolutions to reach its maximum angular velocity. What is the average angular acceleration of the merry-go-round?
- 0.716 rad/s^2
 - 2.05 rad/s^2
 - 3.22 rad/s^2
 - 6.44 rad/s^2
 - 20.3 rad/s^2

- P10. Wind blows across a hollow log that is open on both ends, producing an eerie sound at a frequency of 144 Hz. You measure the log to be 4.76 meters long. Which harmonic is resonating in the log? Assume the air temperature is 20.0°C.
- the $n = 1$ fundamental harmonic
 - the $n = 2$ harmonic
 - the $n = 3$ harmonic
 - the $n = 4$ harmonic
 - the $n = 5$ harmonic
- P11. A cube of aluminum has a mass of 2.80kg. What is the apparent weight of the cube when it is completely submerged in pure water? The density of aluminum is 2700kg/m³.
- 10.2 N
 - 17.3 N
 - 22.4 N
 - 27.4 N
 - 37.6 N
- P12. You explore the electrical properties of dry pasta by connecting one strand of spaghetti to a 24.0V power supply. The current flowing through the spaghetti is measured to be 56.0nA. The cylindrical strand of spaghetti is 36.0cm long and has a diameter of 1.20mm. What is the resistivity of the strand of dry spaghetti?
- 1350 Ωm
 - 2930 Ωm
 - 4300 Ωm
 - 5380 Ωm
 - 7330 Ωm
- P13. Two charges are placed as illustrated. A +400.0nC charge is placed at (0.0, 24.0cm) and a +250.0nC charge is placed at (18.0cm, 0.0). What is the electric potential (voltage) at the origin (0.0, 0.0) due to these two charges?
- 2.50 kV
 - 6.94 kV
 - 9.99 kV
 - 27.5 kV
 - 132 kV
- P14. A particle with the same charge as a proton is moving at a speed of 4.50×10^6 m/s. The particle enters a region where there is a 0.780T magnetic field that is oriented perpendicular to the motion of the particle. The circular path traced out by the particle in the field has a radius of 2.20m. What is the mass of the particle as compared to the mass of a proton?
- $6.11m_p$
 - $18.3m_p$
 - $36.5m_p$
 - $51.8m_p$
 - $67.1m_p$
- P15. An unpolarized beam of light with an initial intensity of 600W/m² passes through a polarizer whose axis is oriented horizontally. After passing through that first polarizer, the beam goes through a second polarizer whose axis is oriented at 30.0° with respect to horizontal. What is the intensity of the beam after passing through both polarizers?
- 75.0 W/m²
 - 150 W/m²
 - 225 W/m²
 - 260 W/m²
 - 450 W/m²
- P16. You look into the still water of a quiet pond and see a minnow swimming just below the surface. The minnow *appears* to be swimming at a point that is 12.0cm deep. How deep below the surface is the minnow *actually* swimming? The index of refraction of water is 1.33.
- 8.00 cm
 - 9.00 cm
 - 12.0 cm
 - 13.3 cm
 - 16.0 cm



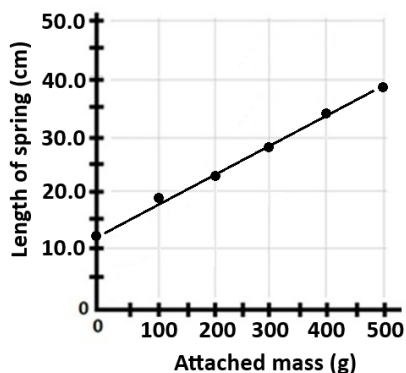
P17. While studying the spectrum of hydrogen, you observe a spectral line with a wavelength of 1005nm. Which transition in hydrogen produces this spectral line?

- A) $n = 6$ to $n = 3$
- B) $n = 7$ to $n = 3$
- C) $n = 7$ to $n = 4$
- D) $n = 8$ to $n = 3$
- E) $n = 8$ to $n = 4$

P18. The K^+ meson has a strangeness of +1, a charge number of +1, and a spin of 0. Its lifetime is $1.2 \times 10^{-8}s$, and it commonly decays into a pair of pions: $K^+ \rightarrow \pi^+ + \pi^0$. What fundamental force is responsible for the decay of the K^+ meson?

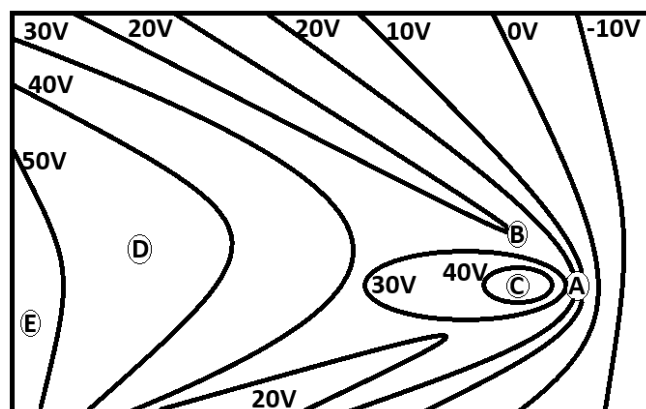
- A) the Weak Force
- B) the Strong Force
- C) the Electromagnetic Force
- D) the Gravitational Force
- E) the Higgs Force

P19. A spring dangles vertically from a hook attached to the ceiling. The unstretched length of the spring is 12.0cm. Various masses are attached to the lower end of the spring, and the length of the spring is measured for each mass. The data are plotted below. Based on these data, what is the approximate spring constant of the spring?



- A) 0.50 N/m
- B) 5.0 N/m
- C) 10 N/m
- D) 13 N/m
- E) 18 N/m

P20. The following map illustrates the equipotential surfaces in a region of space. Based on this map, at what location is the electric field at its greatest magnitude?



- A) location A
- B) location B
- C) location C
- D) location D
- E) location E

Chemistry																18	
1A 1																	2A 2
1 H 1.01																	2 He 4.00
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8 9 10			1B 11	2B 12	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.9	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (289)	116 Lv (293)	117 Ts (293)	118 Og (294)

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Water Data

T_{mp}	= 0°C
T_{bp}	= 100°C
c_{ice}	= 2.09 J/g·K
c_{water}	= 4.184 J/g·K
c_{steam}	= 2.03 J/g·K
ΔH_{fus}	= 334 J/g
ΔH_{vap}	= 2260 J/g
K_f	= 1.86 °C/m
K_b	= 0.512 °C/m

Constants

R	= 0.08206 L·atm/mol·K
R	= 8.314 J/mol·K
R	= 62.36 L·torr/mol·K
e	= 1.602×10^{-19} C
N_A	= 6.022×10^{23} mol ⁻¹
k	= 1.38×10^{-23} J/K
h	= 6.626×10^{-34} J·s
c	= 3.00×10^8 m/s
R_H	= 2.178×10^{-18} J
m_e	= 9.11×10^{-31} kg
$\mathcal{F}$	= 96,485 C/mol e ⁻
1 amp	= 1 C/sec
1 mol e ⁻	= 96,485 C

Densities (g/mL)

Ethanol	0.79
Mineral oil	0.81 – 0.89
Vegetable oil	0.92
Water	1.00
Corn syrup	1.38 – 1.48
Lego piece	1.03 – 1.06

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Physics

Useful Constants

quantity	symbol	value
Free-fall acceleration	g	9.80 m/s^2
Permittivity of Free Space	ϵ_0	$8.854 \times 10^{-12} \text{ C}^2/\text{Nm}^2$
Permeability of Free Space	μ_0	$4\pi \times 10^{-7} \text{ Tm/A}$
Coulomb constant	k	$8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$
Speed of light in a vacuum	c	$3.00 \times 10^8 \text{ m/s}$
Fundamental charge	e	$1.602 \times 10^{-19} \text{ C}$
Planck's constant	h	$6.626 \times 10^{-34} \text{ Js}$
Electron mass	m_e	$9.11 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.67265 \times 10^{-27} \text{ kg}$ 1.007276 amu
Neutron mass	m_n	$1.67495 \times 10^{-27} \text{ kg}$ 1.008665 amu
Atomic Mass Unit	amu	$1.66 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
Gravitational constant	G	$6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$
Stefan-Boltzmann constant	σ	$5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$
Universal gas constant	R	$8.314 \text{ J/mol} \cdot \text{K}$ $0.082057 \text{ L} \cdot \text{atm/mol} \cdot \text{K}$
Boltzmann's constant	k_B	$1.38 \times 10^{-23} \text{ J/K}$
Speed of Sound (at 20°C)	v	343 m/s
Avogadro's number	N_A	$6.022 \times 10^{23} \text{ atoms/mol}$
Electron Volts	eV	$1.602 \times 10^{-19} \text{ J/eV}$
Distance Conversion	miles $\rightarrow$ meters feet $\rightarrow$ meters	1.00 mile = 1609 meters 1.00 foot = 0.3048 meters
Rydberg Constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
Standard Atmospheric Pressure	1 atm	$1.013 \times 10^5 \text{ Pa}$
Density of Pure Water	ρ_{water}	1000.0 kg/m^3

**UIL HIGH SCHOOL SCIENCE CONTEST
ANSWER KEY
2026 INVITATIONAL B**

Biology

B01. C
B02. E
B03. B
B04. A
B05. C
B06. D
B07. A
B08. B
B09. C
B10. E
B11. A
B12. E
B13. D
B14. A
B15. B
B16. B
B17. A
B18. E
B19. B
B20. C

Chemistry

C01. D
C02. B
C03. E
C04. B
C05. C
C06. D
C07. A
C08. C
C09. A
C10. B
C11. E
C12. C
C13. D
C14. A
C15. A
C16. C
C17. A
C18. E
C19. C
C20. B

Physics

P01. E
P02. C
P03. D
P04. E
P05. D
P06. B
P07. B
P08. A
P09. C
P10. D
P11. B
P12. A
P13. D
P14. C
P15. C
P16. E
P17. B
P18. A
P19. E
P20. A

CHEMISTRY SOLUTIONS – UIL INVITATIONAL B 2026

- C01. (D) Potassium carbonate is K_2CO_3 , so there are six atoms.
- C02. (B) The balanced equation is $4 \text{Fe} + 3 \text{O}_2 \rightarrow 2 \text{Fe}_2\text{O}_3$
- C03. (E) Atomic mass is a weighted average of all naturally-occurring isotopes of an element and is the number at the bottom of a periodic table entry. Mass number is the mass of a single atom or a single isotope and is the sum of the protons and neutrons in the nucleus of that atom or isotope.
- C04. (B) A molecule has to have at least one double or triple bond to have π bonds. CH_4 , H_2O , and NH_3 all have only single bonds, while CO_2 has two double bonds and CO has one triple bond.
- C05. (C) $22.4 \text{ g CO}_2 / 44.01 \text{ g/mol} = 0.50898 \text{ mol CO}_2$.
 $PV = nRT$, $V = nRT/P = (0.50898)(0.08206)(298)/(1) = 12.4 \text{ L}$.
- C06. (D) The Lego SCUBA diver is made of ABS plastic which according to the data page has a density of $1.03 - 1.06 \text{ g/cm}^3$. This is greater than the density of water but less than the density of corn syrup, so the diver will sink to the bottom of the water (layer D) but float on the corn syrup (layer E).
- C07. (A) A positive ΔH means a reaction is endothermic and must absorb energy from its surroundings in order for the reaction to proceed.
- C08. (C) An atom is oxidized when its formal charge increases (becomes more positive). In reaction (i), the formal charge on the carbon atom changes from -4 to $+4$, in (ii) it also goes from -4 to $+4$, in (iii) it goes from $+2$ to $+4$, and in (iv) it goes from $+4$ to $+4$ (remains the same).
- C09. (A)
- C10. (B) Only raising the temperature will change K . The other changes will change the amounts of reactant and product present at equilibrium but will not change K itself.
- C11. (E) The p function means “take the $\log_{10}$ of this thing and change the sign,” so for example pH is the negative log of $[\text{H}^+]$. To go the other way, from $\text{p}X$ to just X , $X = 10^{-\text{p}X}$. The p function is commonly used in science to make a set of unwieldy numbers that range between 0 and 1 (such as acid concentrations ranging from 1 M down to 10^{-14} M) fit into an easier numeric range to work with, such as 0-14.
- C12. (C) The Cl^- ion present in solution from the NaCl will inhibit the solubility of the added AgCl due to the common Cl^- ion present from the dissolved NaCl . The other answers are wrong because: A and B: the common ion effect only applies when you have two different species with a common ion, not when you simply have more of the same compound; D) in this case the added NaCl would dissolve and some of the AgCl would precipitate out of solution. This is an example of the common ion effect, but the question specifically asks about affecting the solubility of the *added salt*, and in this case, it is the solubility of the *dissolved salt* that would be affected.

- C13. (D) The integrated rate law for the reaction relates the concentration of one reactant and time. For example, for a zero order reaction is $[A] = [A]_0 - kt$. Where $[A]$ is the current concentration of A , $[A]_0$ is the starting concentration, k is the rate constant for the reaction, and t is time. First order and second order reactions have different integrated rate laws that also relate $[A]$ to $[A]_0$ and t .
- C14. (A) There are two atoms bonded together so this is a molecule, but there is only one type of atom present, so this is an element, not a compound. A compound has to have more than one type of atom in it. This is not a mixture of any kind because all of the particles in the sample are the same. This is a pure diatomic element like N_2 or O_2 .
- C15. (A) For a rigorous solution to this you would create a RICE table and solve the resulting quadratic equation for x , where $x = [OH^-]$. However, when the K_b is small (generally around 10^{-5} or smaller) and the concentration is high (around 0.001 M or higher), x is small compared to the concentration of the base C_b and the quadratic equation simplifies to

$$[OH^-] = \sqrt{C_b K_b}$$

Where C_b = the concentration of the base. As long as $[OH^-] \ll C_b$ (typically 5% or less), the simplified equation can be used. In this case $[OH^-] = (0.75 \text{ M} \times 4.75 \times 10^{-5})^{1/2} = (0.0000356)^{1/2} = 0.0059687 \text{ M} = 0.0060 \text{ M}$ Checking the assumption that $[OH^-] \ll C_b$, $(0.0060/0.75) \times 100 = 0.8\%$, so the simplified equation is good. $[H^+][OH^-] = 1 \times 10^{-14}$, so $[H^+] = 1 \times 10^{-14} / 0.006 = 1.67 \times 10^{-12}$ and $pH = -\log(1.67 \times 10^{-12}) = 11.78$. You do not need the volume of the solution to solve this problem, pH is based only on the H^+ concentration.

- C16. (C) Arrow C is the difference in energy between the reactants and products. If the products have more energy than the reactants, as is the case here, the reaction is endothermic. In an exothermic reaction the products have a lower energy than the reactants.
- C17. (A) A decomposition reaction has one reactant and more than one product. Most decomposition reactions require heat.
- C18. (E) Cu has a molar mass of 63.54 g/mol and Cl has a molar mass of 35.45 g/mol. There are two Cl atoms in $CuCl_2$, so the molar mass of the compound is 134.45. The percent copper is $(63.54/134.45) \times 100\% = 47.3\%$
- C19. (C) $2 NaBrO_3 + Ba(NO_3)_2 \rightarrow Ba(BrO_3)_2 + 2 NaNO_3$ The sodium and nitrate ions are spectators, so the net ionic equation is $Ba^{2+}(aq) + 2 BrO_3^-(aq) \rightarrow Ba(BrO_3)_2(s)$. However, K_{sp} expressions are always written for the dissolution of the solid precipitate, so even though this question asks about forming a precipitate, the K_{sp} expression is still based on the reverse reaction, $Ba(BrO_3)_2(s) \rightarrow Ba^{2+}(aq) + 2 BrO_3^-(aq)$. The equilibrium constant is products over reactants raised to their stoichiometric coefficients, leaving out the solid, so $K_{sp} = [Ba^{2+}][BrO_3^-]^2$. (You can write a K_{eq} expression for the forward reaction which would be answer D, but that K_{eq} would not be K_{sp} .)
- C20. (B) 40 minutes = 2400 seconds. $2 \text{ mA} = 2 \text{ C/second} \times 2400 \text{ sec} = 4800 \text{ C}$. $4800 \text{ C} / 96485 \text{ C/mol} = 0.0497 \text{ mol of electrons} = 0.0497 \text{ mol of Ag} = 0.050 \text{ mol Ag}$.

PHYSICS SOLUTIONS – UIL INVITATIONAL B 2026

- P01. (E) page 63: “Soddy promptly purchased twenty milligrams to use in Ramsay’s laboratory for experiments on radium emanation. Their first attempts to isolate the emanation failed dismally, but spectral tests showed a helium line in the mixture of gases that radium produced.... With this larger sample, Soddy and Ramsay obtained nearly all of helium’s visible spectrum.”
- P02. (C) pages 72-73: “Villard decided to apply the definitive test for charged particles. He sent radium rays through a magnetic field. The magnet broke the beam into two parts. One part was deflected like beta rays. The other part continued in a straight line, oblivious to the magnet. This second portion...traveled through nearly ten inches of air, an aluminum plate, and several pieces of paper....If these powerful rays were neither alpha nor beta rays, what could they be? ... To continue the naming pattern established by Rutherford, these rays were called gamma rays...”
- P03. (D) page 83: “In 1906 physicists Norman Campbell and Albert B. Wood claimed that potassium and rubidium, two metals very different from uranium, radium, and thorium, were weakly radioactive. Tests eliminated possibilities that the effects came from trace impurities.... The radioactivity of potassium and rubidium seemed genuine.” Note: radioactive potassium-40 is very common in nature. Your body contains some, as do all foods that are sources of potassium, such as bananas.
- P04. (E) The cooler the star, the more slowly it uses up its core hydrogen fuel. So, the longest-lived stars are the stars with the lowest temperatures – that is, the class M red stars. Class O stars use up their fuel quickly and remain on the main sequence for only a few million years, Our Sun is a class G yellow star whose lifetime on the main sequence is around 10 billion years. Class M red stars remain on the main sequence for hundreds of billions of years.
- P05. (D) First, let’s convert the distance into miles:

$$d = 120 \text{ yards} * \frac{3 \text{ feet}}{1 \text{ yard}} * \frac{0.3048 \text{ m}}{1 \text{ foot}} * \frac{1 \text{ mile}}{1609 \text{ m}} = 0.068196 \text{ miles.}$$
 Now, we convert the time to hours:

$$t = 12.5 \text{ min} * \frac{1 \text{ hour}}{60 \text{ min}} = 0.20833 \text{ hours.}$$
 Thus, the velocity in miles per hour is

$$v = \frac{d}{t} = \frac{0.068196 \text{ miles}}{0.20833 \text{ hour}} = 0.327 \text{ mph.}$$
- P06. (B) Since we know the time during which the walnut is falling, we can use the kinematic equation:

$$x = x_i + v_i t + \frac{1}{2} a t^2.$$
 The final position is the floor ($x = 0$) and taking downward to be negative, we get: $0 = 2.35 + v_i(0.250) + \frac{1}{2}(-9.80)(0.250)^2 = 2.35 + 0.25v_i - 0.30625.$ This leads to $-2.044 = 0.25v_i \rightarrow v_i = -8.18 \text{ m/s}.$ Thus, the initial velocity was $v_i = 8.18 \text{ m/s}$ downward.
- P07. (B) Since there is no friction, the only two forces acting on the notebook are gravity (mg , directed downward), and the normal force (F_N , directed up and right, perpendicular to the incline). As we usually do when on an incline, we tilt our coordinate system so that the positive x-direction points down and right, parallel to the inclined plane, and the positive y-direction points up and right, perpendicular to the inclined plane. This means that the normal force points entirely in the positive y-direction. The gravitational force is at an angle with respect to our axis system, but it can be broken into components. In the positive x-direction, we have $mg \sin \theta$, and in the negative y-direction we have $mg \cos \theta$. Here $\theta = 36.0^\circ$ is the angle of the incline with respect to horizontal. We have no motion in the y-direction, so all forces in the y-direction must sum to zero. That is $\sum F_y = 0 \rightarrow F_N - mg \cos \theta = 0 \rightarrow F_N = mg \cos \theta.$ Thus, we find that the magnitude of the normal force is $F_N = (1.50 \text{ kg})(9.80 \text{ m/s}^2) \cos(36.0^\circ) = 11.9 \text{ N}.$

- P08. (A) By conservation of momentum, the total momentum of the system prior to the collision must equal the total momentum of the system after the collision. Prior to the collision, the cart has no momentum since it is at rest. The only momentum is that of the clay blob:
 $p_i = m_{cl}v_{ci} = (0.350\text{kg})(13.0\text{m/s}) = 4.55\text{kgm/s}$. After the collision, both the cart and the clay are moving, and since they are stuck together, they must be moving at the same velocity. So, the final momentum is $p_f = m_{cl}v_f + m_{ca}v_f = (0.350)v_f + (0.600)v_f = 0.950v_f$. By conservation of momentum $p_i = p_f \rightarrow 4.55 = 0.950v_f \rightarrow v_f = 4.79\text{m/s}$.
- P09. (C) First, we convert the angular displacement into radians. The merry-go-round takes two complete revolutions to reach maximum velocity. Thus, $\Delta\theta = 2.0\text{rev} * \frac{2\pi\text{rad}}{1\text{rev}} = 4\pi\text{radians}$. We know nothing about the time it takes to reach the maximum velocity, so we'll use the rotational kinematic equation that does not include time: $\omega_f^2 = \omega_i^2 + 2\alpha\Delta\theta$. This gives $(9.00)^2 = (0)^2 + 2\alpha(4\pi)$, and leads to $8\pi\alpha = 81$, resulting in an angular acceleration of $\alpha = \frac{81}{8\pi} = 3.22\text{rad/s}^2$.
- P10. (D) At 20.0°C , the speed of sound is 343m/s . Since we know the frequency and the speed of sound, we can determine the wavelength of the resonant sound by using $f\lambda = v$. This gives $(144\text{Hz})\lambda = 343$, which leads to a wavelength of $\lambda = 2.38\text{m}$. We relate this wavelength to the length of the log by $n\lambda = 2L$, which gives $n(2.38\text{m}) = 2(4.76) \rightarrow n = 4$.
- P11. (B) Based on the density and mass of the cube of aluminum, we can find its volume. $\rho = \frac{m}{V} \rightarrow 2700\text{kg/m}^3 = \frac{2.80\text{kg}}{V} \rightarrow V = 0.001037\text{m}^3$. Now we can determine the buoyant force acting on the cube: $F_B = \rho_{\text{liquid}}Vg = (1000\text{kg/m}^3)(0.001037\text{m}^3)(9.80\text{m/s}^2) = 10.16\text{N}$. The weight of the cube in air is $W_{\text{air}} = mg = (2.80\text{kg})(9.80\text{m/s}^2) = 27.44\text{N}$. Thus, the apparent weight of the cube of aluminum while submerged in water is $W_{\text{water}} = W_{\text{air}} - F_B = 27.44 - 10.16 = 17.3\text{N}$.
- P12. (A) First, the resistance of the pasta can be found from Ohm's Law:
 $V = IR \rightarrow 24.0\text{V} = (56.0 \times 10^{-9}\text{A})R \rightarrow R = 4.286 \times 10^8 \Omega$. Now we find the cross-sectional area of the strand of pasta: $A = \pi r^2 = \pi \left(\frac{D}{2}\right)^2 = \pi \left(\frac{1.20 \times 10^{-3}}{2}\right)^2 = 1.131 \times 10^{-6}\text{m}^2$. Finally, we relate the resistance of the pasta to its resistivity: $R = \frac{\rho L}{A} \rightarrow 4.286 \times 10^8 = \frac{\rho(0.360\text{m})}{1.131 \times 10^{-6}\text{m}^2} \rightarrow \rho = 1346 \approx 1350 \Omega\text{m}$
- P13. (D) Electric potential is, thankfully, not a vector. Thus, we only need to consider the distance of each charge from the origin. The relevant equation for electric potential is $V = \frac{kQ}{r}$ where r is the distance from the charge to the point of interest. For the first charge, which is 400.0nC , the distance from the origin is 24.0cm . So, the potential due to the first charge is $V_1 = \frac{(8.99 \times 10^9)(400.0 \times 10^{-9})}{0.240} = 14980\text{V}$. For the second charge, which is 250.0nC , the distance from the origin is 18.0cm . Thus, the potential due to the second charge is $V_2 = \frac{(8.99 \times 10^9)(250.0 \times 10^{-9})}{0.180} = 12490\text{V}$. Now, we simply add the electric potentials due to each charge to get the total electric potential at the origin:
 $V_{\text{total}} = 14980 + 12490 = 27470\text{V} \approx 27500\text{V} = 27.5\text{kV}$.
- P14. (C) The radius of the circle traced out by a charged particle in a magnetic field is given by $r = \frac{mv}{qB}$. The field is perpendicular to the motion of the particle, and the charge of the particle is the same as a proton ($q = 1.602 \times 10^{-19}\text{C}$). Thus, we have: $2.20 = \frac{m(4.50 \times 10^6)}{(1.602 \times 10^{-19})(0.780)}$. This gives a mass of $m = 6.109 \times 10^{-26}$. Compared to the mass of a proton, this particle's mass is:
 $m = \frac{6.109 \times 10^{-26}}{1.67265 \times 10^{-27}} m_p = 36.5 m_p$.

- P15. (C) When an unpolarized beam of light passes through a polarizer, the beam's intensity is reduced by half due to the elimination of its cross-polarized components. Thus, after the first polarizer, we have $I_1 = \frac{1}{2}I_0 = \frac{1}{2}(600) = 300 \text{ W/m}^2$. The beam is now horizontally polarized. To get through the second polarizer, we rely on Malus' Law: $I_2 = I_1(\cos \theta)^2$ where θ is the angle between the polarization of the beam and the polarizer. In this case $\theta = 30.0^\circ$. So, the intensity after going through the second polarizer is $I_2 = (300)(\cos(30.0))^2 = 225 \text{ W/m}^2$.
- P16. (E) The equation that describes the location of an image created by a refracting surface is $\frac{n_1}{p} + \frac{n_2}{q} = \frac{n_2 - n_1}{R}$. Because the surface of the water is flat, the radius of curvature is effectively infinite. That is $R = \infty$. Thus, the right-side term $\frac{n_2 - n_1}{R} = \frac{n_2 - n_1}{\infty} = 0$. So, for a flat surface we have: $\frac{n_1}{p} + \frac{n_2}{q} = 0$. The object (the minnow) is in the water, so $n_1 = 1.33$ and we see an image from a vantage point in the air, so $n_2 = 1.00$. The image is what we see, which is located 12.0cm below the water's surface. Since it is below the surface, the image location is negative. Thus, $q = -12.0\text{cm}$. Putting it all together, we can find the object location (where the minnow actually is located): $\frac{n_1}{p} + \frac{n_2}{q} = \frac{1.33}{p} + \frac{1.00}{-12.0} = 0 \rightarrow \frac{1.33}{p} = \frac{1}{12}$. This gives an object location of $p = 1.33(12.0) = 16.0\text{cm}$ below the surface.
- P17. (B) First, let's consider the Rydberg formula, which relates spectral line wavelengths to Hydrogen's atomic energy levels: $\frac{1}{\lambda} = R_H \left(\frac{1}{n_B^2} - \frac{1}{n_A^2} \right)$. Here $R_H = 1.0974 \times 10^7 \text{ m}^{-1}$ is the Rydberg constant, n_A is the higher energy level involved and n_B is the lower energy level involved. Putting in the wavelength gives us: $\frac{1}{1005 \times 10^{-9}} = (1.0974 \times 10^7) \left(\frac{1}{n_B^2} - \frac{1}{n_A^2} \right) \rightarrow \frac{1}{n_B^2} - \frac{1}{n_A^2} = 0.0907$. We don't have any more information from the problem itself, but the answer choices all end at either the $n_B = 3$ or the $n_B = 4$ levels. Let's try them both and see what we get for the starting level. For $n_B = 4$ we get $\frac{1}{16} - \frac{1}{n_A^2} = 0.0907 \rightarrow n_A^2 = -35.5$, which gives an undefined answer. So, $n_B = 4$ cannot be the correct final state. Let's consider $n_B = 3$ instead: $\frac{1}{9} - \frac{1}{n_A^2} = 0.0907 \rightarrow n_A^2 = 49.0 \rightarrow n_A = 7$. Thus, we can see that this spectral line wavelength is the result of a transition from the $n_A = 7$ to the $n_B = 3$ energy levels.
- P18. (A) First, we should note that the "Higgs force" isn't a real force. There are only four fundamental forces, listed as choices A through D. So, we can immediately eliminate E as a choice. Second, the gravitational force has not been observed on the subatomic level. It is far too weak and has not yet been connected to any type of particle decay. This eliminates D as a choice. Now we consider the properties of the K^+ meson. Notice that it has strangeness and a relatively long lifetime. The decay products are pions, which are composed of up and down quarks, and (importantly) have zero strangeness. Thus, a quark flavor change must have taken place in order to go from a strange particle to decay products without strangeness. This suggests that the Weak force is involved. The lifetime confirms it. Strong force decays are very fast ($\sim 10^{-23}\text{s}$) and even Electromagnetic force decays are reasonably fast ($\sim 10^{-16}\text{s}$). Weak force decays are the only ones that are as slow as 10^{-8}s . Thus, the Weak force is primarily responsible for the decay of the K^+ meson.

- P19. (E) The important equation that relates the stretching of a spring to the force pulling on the spring is Hooke's Law: $F = -k\Delta x$. The sign indicates that the spring force is opposite to the direction of the stretching, but we don't need to consider directions to solve this problem, so we can ignore it. Noting that the force to stretch the spring is provided by the weight of each mass, we get: $F = mg$. Putting it all together (and removing the unneeded negative sign), we get: $mg = k\Delta x = k(x - x_0)$, where x_0 is the unstretched length of the spring. The plot is of length versus mass, or x versus m . Rearranging, we see: $\frac{g}{k}m = x - x_0 \rightarrow x = \frac{g}{k}m + x_0$. Now we can tell that the slope of the line on the plot must equal $\frac{g}{k}$. To find the slope, we choose two points on the best fit line that are not data points. I shall choose (140g, 20cm) and (325g, 30cm). Converting to proper units gives us the points (0.140kg, 0.20m) and (0.325kg, 0.30m). Now we find the slope using these two points:
- $$slope = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0.30 - 0.20}{0.325 - 0.140} = \frac{0.10}{0.185} = 0.54 \text{m/kg.}$$
- Equating this to the slope formula derived earlier, we obtain: $0.54 = \frac{g}{k} \rightarrow k = \frac{9.80}{0.54} = 18.1 \approx 18 \text{N/m.}$
- P20. (A) The electric field is proportional to the gradient (rate of change over distance) of the potential. So, the more steeply the potential changes, the greater the electric field. On the equipotential map, steep changes in potential appear as equipotential lines that are closely spaced together. The most closely spaced lines are clearly at point A, which is where the electric field magnitude is the greatest. Point B represents a steep turning point of one equipotential, but that is not indicative of a large electric field. Points C and E are local maximums in potential, but they are not near areas of large gradients. Maximum potential does not equate to maximum electric field. Finally, point D is where the lines are farthest apart, representing the area of lowest electric field magnitude.